

The $1/3$ – $2/3$ Conjecture on the Defect Ladder: The Two- and Three-Defect Chain Theorems

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Abstract

A finite poset is a k -defect chain if it becomes a chain after the deletion of k elements; equivalently, its maximum chain omits exactly k elements. We prove the $1/3$ – $2/3$ conjecture for all posets of defect at most three: every such poset that is not itself a chain has an incomparable pair (u, v) with $1/3 \leq \Pr(u \prec v) \leq 2/3$. The class contains posets of width three and four, does not bound the number of incomparabilities per element, and we exhibit members escaping each previously proved class known to us. The two-defect case is proved by an explicit balance lemma for overlapping insertion windows; for it we also prove equality rigidity: every ordinal-sum-indecomposable poset of defect exactly two has a strictly balanced eligible count, and among chain-plus-two-incomparable-defect presentations the unique non-strict member is the three-element poset Q (itself of defect one) — extending Aigner’s width-two equality classification to an infinite family of width-three posets. The three-defect case is proved by contradiction through a rigidity theory of *closed-third avoidance*: if every incomparable orientation probability avoided the closed interval $[N/3, 2N/3]$, the majority relation would be a strict total order, every defect would carry a unique *heavy gap* holding more than a third of the extension mass, and the heavy gaps would align with the defect order. An exhaustive partition of the surviving configurations by defect shape (antichain, V , Λ , Q) and heavy-gap pattern is then closed cell by cell: by Helly-type coincidence and marginal monotonicity arguments, by weight-preserving reflection dualities, by vacuity theorems, and — for the hardest cells — by exhaustive weak-order chamber decompositions carrying exact rational nonnegative-coefficient certificates (169 chambers for the antichain common-heavy inequality; 31 strict and 31 boundary chambers with 62 signed certificates for the tied-siblings V cell and its one-relation corrections). All certificate corpora are exact, machine-checkable, independently replayed proof objects anchored by cryptographic hashes.

1 Introduction

The $1/3$ – $2/3$ conjecture — posed independently by Kislitsyn [1], Fredman [2], and Linial [3] in connection with sorting under partial information — asserts that every finite poset P that is not a chain contains an incomparable pair (u, v) whose orientation probability under the uniform distribution on linear extensions satisfies $1/3 \leq \Pr(u \prec v) \leq 2/3$. Kahn and Saks [4] proved the first uniform positive bound; the strongest general bound is due to Brightwell, Felsner, and Trotter [9]. The conjecture is known for partial orders of width two (Linial [3], sharpened by Sah [16]), height two [8], posets with at most 14 elements [12, 19], posets in which every element is incomparable to at most six others (Peczarski [10]), series-parallel

posets, N -free posets and polytrees (Zaguia [13, 14]), semiorders (Brightwell [7]), several lattice and Young-diagram families (Olson and Sagan [15]), and is preserved by lexicographic sums [17]. Chan and Pak [18] survey the area.

Every finite poset becomes a chain after deleting some number of elements; call the least such number its *defect*. Since the retained chain of a minimal deletion is a maximum chain, the defect of P equals $|P \setminus C|$ for any maximum chain C . The conjecture restricted to defect k interpolates, as k grows, from the merging problems of the classical literature to the full conjecture. Defect one is contained in width two. This paper settles defects two and three completely, and classifies the extremal cases at defect two.

Three subfamilies of the two-defect class were previously known and are not claimed here: members whose two defect windows coincide carry a defect-exchanging automorphism, a case settled by Peczarski's automorphism theorem [11] (and, for cycle-free members, already by Ganter, Häfner, and Poguntke [6], who obtain a $1/2$ -balanced pair); members of width two are Linial–Sah; members whose windows have length at most two are semiorders. The class as a whole — with windows of unbounded length, hence unboundedly many incomparabilities at a single element, and with genuine width-three members — is contained in neither the width-two class nor any bounded-incomparability class, and its non-containment in the remaining listed classes is witnessed explicitly. With $m = 2$ and windows $I = [0, 1]$, $J = [1, 2]$, the four elements (x, c_1, c_2, y) induce an N ($x < c_2$, $c_1 < c_2$, $c_1 < y$, all other pairs incomparable), both as an induced suborder and in the Hasse diagram — so the class is contained in neither the series-parallel nor the N -free class. With $m = 3$ and windows $I = [1, 2]$, $J = [1, 3]$, the Hasse diagram contains the cycle $c_1 x c_3 c_2$ — so it is not contained in the forest-cover class. Unbounded windows exclude semiorders, and height excludes the height-two class. The incomparable-defect members are precisely the interval orders with exactly two nontrivial intervals; general interval orders remain open.

Section 2 constructs the insertion-window model and its state bijection; Section 3 proves the balance lemma; Section 4 assembles the Two-Defect Chain Theorem; Section 5 proves the equality rigidity classification; Sections 6–7 enter the three-defect ladder through its exactly solvable chambers (the aligned Q chamber, exactly at the boundary, and the adjacent one-endpoint chambers). Sections 8–13 prove the Three-Defect Chain Theorem: Section 8 develops the closed-third avoidance rigidity theory (strong order, heavy gaps, alignment), Section 9 the weighted gap-fiber model and the structural reductions, Sections 10–12 close the twelve shape-by-heavy-pattern cells (antichain; V and Λ ; Q), and Section 13 assembles the theorem. Section 14 documents the certificate corpora as verifiable proof objects, and Section 15 discusses the ladder ahead. Throughout, $e(P)$ is the number of linear extensions of P , $\Pr(u \prec v)$ the fraction orienting u before v , and $\#L(P; \cdot)$ the number of extensions with the displayed property; windows are integer intervals of insertion gaps against the fixed chain C .

2 The doubled-diagonal model

Let $C = (c_1 < \dots < c_m)$ be a chain and let x, y be two further elements with $x \parallel y$. Consistency with the chain forces the admissible insertion gaps of x to form a window $I = [a, b] \subseteq \{0, \dots, m\}$ (gap i meaning that exactly i chain elements precede x): the relations are $c_t < x$ for $t \leq a$ and $x < c_t$ for $t > b$. Similarly y carries a window $J = [c, d]$.

Lemma 2.1 (Overlap). *If $x \parallel y$ then $I \cap J \neq \emptyset$. Conversely, for any two nonempty overlapping windows I, J , imposing the chain order, the threshold relations of both windows, and no relation between x and y defines a valid poset in which $x \parallel y$ and no hidden comparabilities arise.*

Proof. If $b < c$ then $x < c_{b+1} \leq c_c < y$ by transitivity, contradicting $x \parallel y$; the case $d < a$ is symmetric. For the converse, every pair of gaps $(i, j) \in I \times J$ is realized by merging x and y into the chain at those gaps, and at a common gap both relative orders of x, y occur, so $x \parallel y$ and the prescribed windows are exactly attained. $\square$

Lemma 2.2 (State bijection and counts). *Let $p = |I|$, $q = |J|$, $K = I \cap J$, $h = |K|$. The linear extensions of P are in bijection with the states $(i, j) \in I \times J$ with $i \neq j$, together with two ordered states for each common gap $i = j \in K$; hence*

$$N = e(P) = (pq - h) + 2h = pq + h.$$

The orientation counts of the incomparable pairs are

$$\begin{aligned} D &= \#\{(i, j) \in I \times J : i < j\} + h && \text{for the pair } (x, y), \\ A_t &= q \#\{i \in I : i < t\} + \#\{i \in K : i < t\} && \text{for } (x, c_t), \ a < t \leq b, \\ B_t &= p \#\{j \in J : j < t\} + \#\{j \in K : j < t\} && \text{for } (y, c_t), \ c < t \leq d, \end{aligned}$$

and exactly the displayed thresholds correspond to incomparable pairs; outside them the orientation is forced.

Proof. A linear extension is determined by the pair of gaps, except that when x and y occupy a common gap the two relative orders give two extensions — the doubled diagonal. For D : every off-diagonal state with $i < j$ contributes, and exactly the x -before- y diagonal state contributes, giving the displayed count. For A_t with $a < t \leq b$: $x < c_t$ exactly when the x -gap satisfies $i < t$; each such i pairs with q choices of j , plus one extra state when $i \in K$ (the doubled diagonal), and symmetrically for B_t . Incomparability of the displayed pairs is witnessed by explicit gap choices realizing both orientations. $\square$

3 The balance lemma

Theorem 3.1 (Overlapping Two-Interval Balance Lemma). *For every pair of nonempty overlapping windows, at least one of D , an eligible A_t , or an eligible B_t lies in $[N/3, 2N/3]$.*

The proof rests on one discrete intermediate-value fact and exact jump accounting.

Lemma 3.2 (Discrete crossing lemma). *Let $x_1 < \dots < x_r$ be integers with $x_1 \leq N/3$, $x_r \geq 2N/3$, and $x_{k+1} - x_k \leq N/3$ for all k . Then some $x_k \in [N/3, 2N/3]$.*

Proof. Take the first k with $x_k \geq N/3$. If $k = 1$ then $x_1 = N/3$ exactly and we are done; otherwise $x_{k-1} < N/3$, so $x_k = x_{k-1} + (x_k - x_{k-1}) < N/3 + N/3 = 2N/3$. $\square$

Exchanging the defect names replaces D by $N - D$ and swaps the A and B families; translation and order reversal also preserve the assertion. After these symmetries exactly two canonical forms remain: *crossing* ($a \leq c \leq b \leq d$; then $h = p - s$ where $s = c$ after translating a to 0, and $q \geq h$) and *nested* ($J \subseteq I$; then $h = q$).

The crossing form. Direct summation gives

$$N = pq + h, \quad D = pq - \frac{h(h-1)}{2}, \quad A_t = qt + \max(0, t - s), \quad B_{s+u} = pu + \min(u, h),$$

with exact consecutive jumps $A_{t+1} - A_t \in \{q, q+1\}$ and $B_{s+u+1} - B_{s+u} \in \{p, p+1\}$. If $N \geq 3(q+1)$ then $p \geq 2$, the A -sequence runs from $A_1 \leq q+1 \leq N/3$ to $A_{p-1} = N - (q+1) \geq 2N/3$ with jumps at most $q+1 \leq N/3$, and Lemma 3.2 produces a balanced eligible A_t . Symmetrically, if $N \geq 3(p+1)$ the B -sequence does the same. If neither large-case condition holds, the inequalities $q(p-3) < 3-h$ and $p(q-3) < 3-h$ with $1 \leq h \leq \min(p, q)$ force exactly fifteen triples (p, q, h) , each of which carries an explicit balanced witness:

(p, q, h)	s	N	witness
(1, 1, 1)	0	2	$D = 1$
(1, 2, 1)	0	3	$D = 2$
(1, 3, 1)	0	4	$B_{s+1} = 2$
(1, 4, 1)	0	5	$B_{s+1} = 2$
(2, 1, 1)	1	3	$D = 2$
(2, 2, 1)	1	5	$A_1 = 2$
(2, 2, 2)	0	6	$D = 3$
(2, 3, 1)	1	7	$A_1 = 3$
(2, 3, 2)	0	8	$D = 5$
(3, 1, 1)	2	4	$A_2 = 2$
(3, 2, 1)	2	7	$A_2 = 4$
(3, 2, 2)	1	8	$D = 5$
(3, 3, 1)	2	10	$A_2 = 6$
(3, 3, 2)	1	11	$A_2 = 7$
(4, 1, 1)	3	5	$A_2 = 2$

The nested form. Here $N = (p+1)q$. If $q \geq 2$, take $u = \lceil q/3 \rceil$: then $1 \leq u \leq q-1$ and $q \leq 3u \leq 2q$, so the eligible count $B_{s+u} = (p+1)u$ satisfies $N \leq 3B_{s+u} \leq 2N$ *exactly* — the factorization of N makes the nested case rigid. If $q = 1$ then $N = p+1$, $D = s+1$, and $A_t = t + \mathbf{1}_{\{s < t\}}$; for $p \geq 5$ the A -sequence starts at most 2, ends at least $p-1$, and jumps by 1 or 2, so Lemma 3.2 applies, while $p \leq 4$ is a finite table of explicit witnesses.

Remark 3.3 (D alone does not suffice). The count D alone is not always balanced: at $(p, q, h) = (2, 2, 1)$ one has $N = 5$ and $D = 4 > 2N/3$, while the eligible $A_1 = 2$ is balanced. Any proof must use the chain-cut counts.

4 The two-defect chain theorem

Theorem 4.1 (Two-Defect Chain Theorem). *Every finite non-chain poset P containing a chain C with $|P \setminus C| = 2$ has an incomparable pair (u, v) with $1/3 \leq \Pr(u \prec v) \leq 2/3$.*

Proof. Write $P \setminus C = \{x, y\}$. If x and y are comparable, then $C \cup \{x < y\}$ is a two-chain cover, so P has width at most two and the theorem of Linial [3] (in the sharpened form of Sah [16]) applies. If $x \parallel y$, their windows overlap by Lemma 2.1; Lemma 2.2 identifies the orientation counts of the genuine incomparable pairs, and Theorem 3.1 produces one in the middle third. $\square$

5 The extremal classification

Write Q for the three-element poset with exactly one relation. Aigner [5] proved that the width-two posets with balance constant exactly $1/3$ are precisely the ordinal sums of singletons and copies of Q . The census of Gupta [19] determines the extremal balance data through order 14: exactly 128 classes attain $1/3$, every one an ordinal sum of singletons and copies of Q — in particular all of width two. The following theorem proves this non-attainment for the full (infinite, width-three-containing) class. Note the distinction between the *presentation* (a chain plus two designated defects, the model of Section 2, which any poset of defect at most two admits) and the *exact defect* $n - h(P)$ of the abstract: the extremal member Q arises as a one-element chain plus two defects but has exact defect one.

Theorem 5.1 (Equality rigidity). (a) *Every ordinal-sum-indecomposable poset of defect exactly two has an eligible orientation count strictly inside the open interval $(N/3, 2N/3)$; in particular its balance constant exceeds $1/3$.* (b) *Among ordinal-sum-indecomposable posets presented as a chain plus two incomparable defects, the unique member all of whose eligible counts avoid the open interval is Q — which has defect one. Every other member has an eligible count strictly inside.*

The proof classifies the *simultaneous avoiders* — parameter tuples for which every eligible count C satisfies $3C \leq N$ or $3C \geq 2N$ — and intersects them with an indecomposability criterion:

Proposition 5.2 (Avoider classification). *In the canonical crossing form, simultaneous avoidance holds exactly for $(p, q, h, s) = (1, 2, 1, 0)$ and $(2, 1, 1, 1)$. In the nested form, exactly for $(p, q, s) = (2, 1, 0)$ and $(2, 1, 1)$.*

Proposition 5.3 (Indecomposability criterion). *The two-defect model with windows $I = [a, b]$, $J = [c, d]$ is ordinal-sum-indecomposable iff its incomparability graph is connected, iff every $t \in \{1, \dots, m\}$ satisfies $a < t \leq b$ or $c < t \leq d$; under the overlap hypothesis this is equivalent to $\min(a, c) = 0$ and $\max(b, d) = m$.*

Proof of Proposition 5.2. Write $\text{AV}(C)$ for the condition $3C \leq N$ or $3C \geq 2N$. We first sharpen Lemma 3.2 under avoidance: if $x_1 \leq N/3$, $x_r \geq 2N/3$, all jumps are at most $N/3$, and every term satisfies AV, then the first term $\geq N/3$ equals $N/3$ exactly, and its successor equals $2N/3$ exactly (the predecessor of the first crossing term is strictly below $N/3$, so the jump bound puts that term strictly below $2N/3$; avoidance then forces equality at $N/3$, and the next term can neither be in the open middle nor overshoot $2N/3$).

Large regimes. In the crossing form the A -sequence has $A_1 \leq q + 1$, $A_{p-1} = N - (q + 1)$, and jumps in $\{q, q + 1\}$. If $N \geq 3(q + 1)$, all- A avoidance forces $N/3 \in \{q, q + 1\}$ by the sharpened lemma, hence $N = 3(q + 1)$, i.e. $q(p - 3) = 3 - h$. With $1 \leq h \leq \min(p, q)$ the solutions are $(p, q, h) = (3, q, 3)$ for $q \geq 3$, $(4, 2, 1)$, and $(5, 1, 1)$; the first is killed at $q = 3$ by $D = 6$, $N = 12$ (strict middle) and for $q > 3$ by the symmetric B -argument (which would force $N = 3(p + 1) = 12$, impossible), while the last two carry the strict witnesses $A_2 = 4$, $N = 9$ and $A_3 = 3$, $N = 6$. Symmetrically, $N \geq 3(p + 1)$ with all- B avoidance forces $p(q - 3) = 3 - h$, whose solutions $(p, 3, 3)_{p \geq 3}$, $(2, 4, 1)$, $(1, 5, 1)$ are killed the same way. Hence every survivor satisfies $N < 3(q + 1)$ and $N < 3(p + 1)$.

Residual list. The strict inequalities $q(p-3) < 3-h$ and $p(q-3) < 3-h$ force $p, q \leq 4$ and exactly the fifteen triples of the table in Section 3.1. All but two carry an eligible count strictly inside the open middle third:

$(1, 1, 1): D=1, N=2$	$(1, 3, 1): B_1=2, N=4$	$(1, 4, 1): B_1=2, N=5$
$(2, 2, 1): A_1=2, N=5$	$(2, 2, 2): D=3, N=6$	$(2, 3, 1): A_1=3, N=7$
$(2, 3, 2): D=5, N=8$	$(3, 1, 1): A_2=2, N=4$	$(3, 2, 1): A_2=4, N=7$
$(3, 2, 2): D=5, N=8$	$(3, 3, 1): A_2=6, N=10$	$(3, 3, 2): A_2=7, N=11$
$(4, 1, 1): A_2=2, N=5$		

(each displayed value C satisfies $N < 3C < 2N$). The remaining two triples avoid: at $(1, 2, 1)$, $N = 3$ with $D = B_1 = 2$, both at the boundary $3C = 2N$; at $(2, 1, 1)$, $N = 3$ with $D = 2$ and $A_1 = 1$, at the boundaries $3C = 2N$ and $3C = N$. Note $D - N/2 = (pq - h^2)/2 \geq 0$ throughout, so D never realizes the low branch in this orientation.

Nested form. Here $N = (p+1)q$ and $B_{s+u} = (p+1)u$. If $q \geq 2$ and $3 \nmid q$, then $u = \lceil q/3 \rceil$ gives $q < 3u < 2q$, a strict-middle B ; if $q = 3k \geq 6$, take $u = k + 1$. If $q = 3$: $p = 3$ gives $D = 6$, $N = 12$ strict middle, and for $p \geq 4$ the A -sequence starts at most $4 < N/3 = p + 1$, ends at least $3p - 1 > 2N/3$, with jumps at most 4, so Lemma 3.2 gives a strict term. For $q = 1$: $N = p + 1$ and $\{D\} \cup \{A_t\} = \{1, \dots, p\}$; $p = 1$ fails at $D = 1$, $N = 2$, $p \geq 3$ contains an integer strictly inside, and $p = 2$ survives for $s \in \{0, 1\}$ with values $\{1, 2\}$ at the boundaries of $N = 3$. $\square$

Proof of Proposition 5.3. The incomparability graph has the edge xy , no edges among chain vertices, and c_t adjacent to x exactly for $a < t \leq b$, to y exactly for $c < t \leq d$. After exchange assume $a \leq c$; overlap gives $c \leq b$, so the union of the two adjacency windows is the single index interval $(a, \max(b, d)]$, and the graph is connected exactly when that interval is all of $\{1, \dots, m\}$ — that is, $\min(a, c) = 0$ and $\max(b, d) = m$. Connectedness of the incomparability graph is equivalent to ordinal-sum indecomposability, since the canonical ordinal-sum factors are precisely the incomparability components. $\square$

Intersecting the avoider tuples (with their translations, defect exchanges, and order reversals) with the criterion leaves exactly the four interval presentations (L, F) , (F, L) , (F, R) , (R, F) , where $L = [0, 0]$, $R = [1, 1]$, $F = [0, 1]$ at $m = 1$: endpoint coverage forces $m = s + q - 1$, collapsing both survivor tuples to one-element chains with the displayed windows, each isomorphic to Q (extensions abc, acb, cab ; pair counts $(2, 1)$ and $(1, 2)$). Every other arithmetic survivor has a nontrivial chain prefix or suffix and is a chain- Q -chain ordinal sum, hence decomposable. This proves part (b) of Theorem 5.1.

Part (a) follows. Let P be indecomposable of defect exactly two, and let C be a maximum chain, $P \setminus C = \{x, y\}$. If x and y are comparable, $C \cup \{x < y\}$ is a two-chain cover, so P has width two; Linial [3] gives $\delta(P) \geq 1/3$, and by Aigner's classification [5] equality holds only for ordinal sums of singletons and copies of Q — all decomposable except Q itself, whose defect is one. Hence $\delta(P) > 1/3$: some incomparable pair lies strictly inside. If $x \parallel y$, part (b) applies to the presentation, and its unique avoider Q has defect one, not two; so P carries a strict eligible count.

Remark 5.4. An eligible count with $N/3 < C < 2N/3$ yields a strict witness; if all eligible counts avoid the open interval, Theorem 3.1 forces equality at $N/3$ or $2N/3$ — which is the

situation classified above. A corroborating exact scan through $p, q \leq 60$ (73,810 canonical cases) found no further avoiders.

6 Three defects: the aligned chamber

We now take three defects with the induced order Q itself: $x < y$, and z incomparable to both. Suppose first that all three insertion windows coincide in $I = [a, b]$ with $p = |I| \geq 2$.

Lemma 6.1 (Triangular weighted-state model). *The linear extensions are represented exactly by the triples $(i, j, k) \in I^3$ with $i \leq j$ (i, j, k the gaps of x, y, z), with multiplicity 1 when the gaps are distinct or $i = j \neq k$, multiplicity 2 when k ties exactly one of i, j , and multiplicity 3 when $i = j = k$. In particular*

$$N = e(P) = \frac{p(p+1)(p+2)}{2}.$$

Theorem 6.2 (Aligned Q -Defect Chain Theorem). *In the aligned model, $\Pr(x \prec z) = \Pr(z \prec y) = 2/3$ and $\Pr(z \prec x) = \Pr(y \prec z) = 1/3$ exactly.*

Proof (orbit argument). Delete the relation $x < y$. The three defects become interchangeable, and relabeling symmetry partitions the extensions of the weakened poset equally among the six relative orders of $\{x, y, z\}$, each class of size $\binom{p+2}{3}$. Restoring $x < y$ retains exactly the classes xyz, xzy, zxy . Two of the three have $x \prec z$, and two have $z \prec y$; the counts are exact thirds of $N = 3\binom{p+2}{3}$. $\square$

Proposition 6.3 (Chain-cut counts). *For an eligible cut c_t with $r = t - a \in \{1, \dots, p-1\}$, set $H_1 = \binom{p+2}{3} - \binom{p-r+2}{3}$, $H_2 = r(r+1)(3p-2r+2)/6$, $H_3 = \binom{r+2}{3}$. Then $\#L(x < c_t) = 2H_1 + H_2$, $\#L(y < c_t) = H_2 + 2H_3$, $\#L(z < c_t) = H_1 + H_2 + H_3$, with complements N - these; outside the eligible interval the orientation is forced.*

Remark 6.4. The aligned family generalizes Q ($p = 2$, $m = 1$ yields Q together with one isolated element — the chain element c_1 , incomparable to all three defects) and sits exactly on the conjecture's boundary at its defect pairs for *every* window size — an infinite three-defect family of boundary-exact pairs. Whether any member's balance *constant* equals $1/3$ is determined by Proposition 6.3 and left to the discussion.

7 Three defects: the adjacent one-endpoint chambers

Now break the symmetry minimally: keep the windows of x, y equal to I ($p \geq 2$) and let the window J of z be I with one endpoint gap added or deleted.

Lemma 7.1 (Feasible states). *In every admissible adjacent one-endpoint chamber the linear extensions are exactly the triples (i, j, k) with $i, j \in I$, $i \leq j$, $k \in J$, with weight $w(i, j, k) = 1 + \mathbf{1}_{\{k=i\}} + \mathbf{1}_{\{k=j\}}$. With $M = \binom{p+2}{3}$, $T = \binom{p+1}{2}$, $S = \binom{p+2}{2}$, the totals are*

$$N_+ = 3M + T = \frac{p(p+1)(p+3)}{2} \quad (\text{endpoint addition}), \quad N_- = 3M - S = \frac{(p-1)(p+1)(p+2)}{2} \quad (\text{endpoint deletion})$$

Theorem 7.2 (Adjacent One-Endpoint Q Balance Theorem). *Every admissible adjacent one-endpoint chamber has a balanced incomparable defect pair. Explicitly: right endpoint addition admits the witness $\{y, z\}$ with $\Pr = \frac{p+5}{3(p+3)}$; left endpoint addition admits $\{x, z\}$ with $\Pr = \frac{2(p+2)}{3(p+3)}$; right endpoint deletion admits $\{x, z\}$ with $\Pr = \frac{p(2p+5)}{3(p+1)(p+2)}$; left endpoint deletion admits $\{y, z\}$ with $\Pr = \frac{p^2+4p+6}{3(p+1)(p+2)}$. Each ratio lies in $[1/3, 2/3]$ for every $p \geq 2$.*

Proof. Write $R_x = \#L(z \prec x)$ and $R_y = \#L(z \prec y)$. Summing the weighted states of Lemma 7.1 fiber by fiber (a tied gap $k = i$ contributes both tied orders to $z \prec y$; the triple diagonal contributes its two z -before orders to R_y and one to R_x) gives

$$R_x = \sum_{i=0}^{p-1} (p-i) \#\{k \in J : k < i\} + \sum_{i \in K} (p-i), \quad R_y = \sum_{j=0}^{p-1} (j+1) \#\{k \in J : k < j\} + \sum_{i \in K} (p-i-1) + \sum_{j \in K} (j+1) + h.$$

Evaluating these sums in each chamber, with

$$\begin{aligned} A_0 &= \frac{p(p+1)(p+2)}{6}, & B_0 &= \frac{p(p+1)(p+5)}{6}, & C_0 &= \frac{p(p+1)(2p+7)}{6}, & D_0 &= \frac{p(p-1)(p+1)}{6}, \\ E_0 &= \frac{p(p-1)(2p+5)}{6}, & F_0 &= \frac{(p-1)(p^2+4p+6)}{6}, & G_0 &= \frac{(p-1)(p+1)(p+3)}{3}, \end{aligned}$$

yields the complete orientation table (each orientation pair summing to N):

chamber	N	$\#L(x \prec z)$	$\#L(z \prec x)$	$\#L(y \prec z)$	$\#L(z \prec y)$
right addition	N_+	C_0	A_0	B_0	$2A_0$
left addition	N_+	$2A_0$	B_0	A_0	C_0
right deletion	N_-	E_0	F_0	D_0	G_0
left deletion	N_-	G_0	D_0	F_0	E_0

The displayed witnesses read off the table: $B_0/N_+ = \frac{p+5}{3(p+3)}$ (balanced since $p+5 \geq p+3$ and $p+5 \leq 2p+6$); $2A_0/N_+ = \frac{2(p+2)}{3(p+3)}$ (since $2(p+2) \geq p+3$ and $2(p+2) \leq 2(p+3)$); $E_0/N_- = \frac{p(2p+5)}{3(p+1)(p+2)}$ (lower bound $\iff p^2+2p-2 \geq 0$, upper $\iff p+4 \geq 0$); and $F_0/N_- = \frac{p^2+4p+6}{3(p+1)(p+2)}$ (the same two inequalities exchanged). All hold for $p \geq 2$; reverse orientations are complements, hence balanced as well. $\square$

This is the first chamber in which the triple-diagonal machinery operates without full relabeling symmetry: the orbit argument no longer applies, and balance is carried by the explicit fixed witnesses above.

8 Closed-third avoidance: the rigidity theory

From here on, the argument is a proof by contradiction, and this section builds the contradiction hypothesis into a rigid structure. Say that P satisfies *closed-third avoidance* if

$$\Pr(u \prec v) \notin [1/3, 2/3] \quad \text{for every orientation of every incomparable pair } \{u, v\}.$$

The Three-Defect Chain Theorem is exactly the statement that no ordinal-sum-indecomposable non-chain poset with $|P \setminus C| \leq 3$ satisfies it.

Theorem 8.1 (Strong order). *Under closed-third avoidance, the relation $u \prec v \iff \Pr(u \prec v) > 2/3$ is a strict total order on P extending the poset order. If P is not a chain, some adjacent pair of this total order is incomparable in P .*

Proof. Totality: for comparable pairs the orientation probability is 1; for incomparable pairs, $\Pr(u \prec v) + \Pr(v \prec u) = 1$, and avoidance places exactly one of the two above $2/3$. Transitivity: if $A = \{L : x <_L y\}$ and $B = \{L : y <_L z\}$ have $|A|, |B| > 2N/3$, then $|A \cap B| \geq |A| + |B| - N > N/3$, and every extension in $A \cap B$ has $x <_L z$; so $\Pr(x \prec z) > 1/3$. If $z <_P x$ this event would be empty, a contradiction, so either $x <_P z$ (whence $x \prec z$ directly) or $x \parallel z$, in which case avoidance excludes $[1/3, 2/3]$ and the established lower bound forces $\Pr(x \prec z) > 2/3$. For the adjacent pair: if every adjacent pair of the total order were comparable in P , transitivity of P would make every pair comparable, so P would be a chain. $\square$

Remark 8.2 (The closed interval is exact). The open-interval analogue of avoidance is attainable: the four-element poset $r < a < b, r < c$ has three extensions and its two incomparable pairs sit exactly at $1/3$ and $2/3$. The conjecture's closed target $[1/3, 2/3]$ is therefore the correct boundary convention, and it is the closed hypothesis that the assembly refutes.

For a defect $u \notin C$ and an extension L , let $G_u(L) \in \{0, \dots, m\}$ be the *gap* of u : the number of chain elements preceding u in L . Legality confines G_u to the window $I_u = [a_u, b_u]$, and the chain-cut events are gap events: $\{L : u <_L c_t\} = \{G_u < t\}$. Write $\mu_u(r) = \#\{L : G_u(L) = r\}$.

Lemma 8.3 (Nonsingleton windows). *If C is a maximum chain and $u \notin C$, then $|I_u| \geq 2$.*

Proof. If $I_u = \{r\}$, then u is comparable with every chain element and inserts at one position, so $C \cup \{u\}$ is a chain of size $m + 1$, contradicting maximality. $\square$

Theorem 8.4 (Heavy-gap rigidity). *Under closed-third avoidance, every defect u has a unique gap ρ_u with $\mu_u(\rho_u) > N/3$, and both strict tails satisfy $\sum_{r < \rho_u} \mu_u(r) < N/3$ and $\sum_{r > \rho_u} \mu_u(r) < N/3$.*

Proof. The cumulative counts $F(t) = \#\{L : G_u < t\}$ run from 0 to N as t crosses the window; by Lemma 8.3 the window has interior thresholds, and each interior value $F(t)$ is the orientation count of a genuine incomparable pair $\{u, c_t\}$, hence avoids $[N/3, 2N/3]$. Let t be the last threshold with $F(t) < N/3$ (window endpoints supply $F = 0$). The next value cannot lie in the forbidden closed interval, so $F(t + 1) > 2N/3$, and the single gap $\rho_u = t$ between them has $\mu_u(\rho_u) = F(t + 1) - F(t) > N/3$. Its lower tail is $F(t) < N/3$ and its upper tail is $N - F(t + 1) < N/3$; any other gap sits inside one of the tails, so no second gap reaches $N/3$. Endpoint heavy gaps use the empty outside tail. $\square$

Theorem 8.5 (Heavy-gap alignment). *Let $R = P[P \setminus C]$ be the induced defect order and let u, v be defects with $\rho_u < \rho_v$. Under closed-third avoidance: (a) $v <_R u$ is impossible; (b) if $u \parallel_R v$ then $\Pr(u \prec v) > 2/3$.*

Proof. (a) If $v <_R u$, isotonicity gives $G_v \leq G_u$ in every extension, so $\{G_v = \rho_v\} \subseteq \{G_u \geq \rho_v\} \subseteq \{G_u > \rho_u\}$, whence $\mu_v(\rho_v) \leq \sum_{r > \rho_u} \mu_u(r) < N/3$, contradicting heaviness. (b) On the complement of $\{G_u > \rho_u\} \cup \{G_v < \rho_v\}$ one has $G_u \leq \rho_u < \rho_v \leq G_v$, a strict gap inequality, which forces $u <_L v$. Hence $\Pr(v \prec u) < 1/3 + 1/3 = 2/3$; avoidance excludes $[1/3, 2/3]$, so $\Pr(v \prec u) < 1/3$ and $\Pr(u \prec v) > 2/3$. $\square$

9 The three-defect model and the case grid

Now let $|P \setminus C| = 3$. A *gap map* assigns to each defect u a gap $g(u) \in I_u$, isotone with respect to R ; extensions decompose into fibers over feasible gap maps, and with $B_r = g^{-1}(r)$ the fiber weight is $W(g) = \prod_r e(P[B_r])$, so that $N = \sum_g W(g)$. The block factors here are posets on at most three points, so $W \in \{1, 2, 3, 6\}$; the weighted diagonals of Sections 2 and 6 are the two- and three-defect instances of this formula.

Proposition 9.1 (Shape classification). *Up to order duality, the induced defect order R on three elements is one of: the antichain A ; the one-relation poset Q ($x < y$, z incomparable to both); the shape V ($p < q$, $p < r$, $q \parallel r$) or its dual Λ ; or the three-chain T .*

Proposition 9.2 (Three-chain reduction). *If R is the chain $x < y < z$, then $C \cup \{x < y < z\}$ is a two-chain cover of P , so P has width at most two and the width-two theorem of Linial [3] (sharpened by Sah [16]; the direct-sum exceptional family there is excluded by ordinal-sum indecomposability) supplies a balanced pair.*

By Theorem 8.4 each defect carries a unique heavy gap; three values realize exactly one of three patterns: *all-distinct* (D), *exactly two equal* (2E), or *all equal* (CH). Together with Proposition 9.1 this yields the twelve-cell grid

$$\{A, V, \Lambda, Q\} \times \{D, 2E, CH\},$$

and the Three-Defect Chain Theorem is proved by showing every cell is empty under closed-third avoidance. Sections 10–12 do this shape by shape. We will also use:

Theorem 9.3 (Canonical factorization; balance under ordinal sums). *Every finite nonempty poset factors uniquely as the ordinal sum of the induced subposets on the connected components of its incomparability graph, each factor ordinal-sum-indecomposable; and a uniformly random extension of the sum restricts to independent uniform extensions of the factors, so every incomparable pair has the same orientation probabilities in the sum as in its factor.*

10 The antichain shape

Theorem 10.1 (Heavy-gap coincidence for antichain defects). *Let x, y, z be pairwise incomparable defects. Under closed-third avoidance, $\rho_x = \rho_y = \rho_z$.*

Proof sketch from the banked derivation. With no relations among the defects, the feasible gap maps form the product $I_x \times I_y \times I_z$, and the exact tied-fiber weights make each defect's marginal μ_u largest precisely on the common intersection $K = I_x \cap I_y \cap I_z$: a common gap acquires the tied-block multiplicities, and moving off K strictly sheds them. Pairwise overlap of the three windows (Lemma 2.1 applied pairwise) gives $K \neq \emptyset$ by Helly's theorem in one dimension. Heavy-gap uniqueness (Theorem 8.4) then pins all three heavy gaps to the same point of K . $\square$

This empties A-D and A-2E simultaneously — any non-coincident pattern is impossible. The surviving cell is the common-heavy antichain, which is destroyed by an exact inequality. Translate the common heavy gap to 0, write $I_u = [-a_u, b_u]$, and let T_σ count extensions

by the relative order σ of (x, y, z) (indices 1, 2, 3), so $N = \sum_{\sigma} T_{\sigma}$. Under avoidance the strong order (Theorem 8.1) orients the three defect pairs with probabilities $> 2/3$; labeling so that $x \prec y \prec z$, adding the three inequalities $\#(x < y), \#(y < z), \#(x < z) > 2N/3$ and subtracting $2N$ gives

$$Q_A := T_{123} - T_{231} - T_{312} - 2T_{321} > 0.$$

Meanwhile L_1 (the lower tail of the lowest window's defect) and H_3 (the upper tail of the highest) are strict heavy-gap tails, so $3L_1 < N$ and $3H_3 < N$.

Theorem 10.2 (Centered three-interval CH inequality). *For every nonnegative integer endpoint vector with $I_u = [-a_u, b_u]$: if $3L_1 < N$ and $3H_3 < N$, then $Q_A \leq 0$.*

Proof by certificate corpus. Set $S_L = N - 3L_1$, $S_H = N - 3H_3$. The weak orderings of the two endpoint triples (a_1, a_2, a_3) and (b_1, b_2, b_3) partition parameter space into $13 \times 13 = 169$ chambers, on each of which N, L_1, H_3, Q_A are polynomials in the chamber's nonnegative coordinates. For 168 chambers there are exact rational $\alpha, \beta \geq 0$ with

$$-Q_A = \alpha S_L + \beta S_H + \text{Rem},$$

where every coefficient of Rem is nonnegative — so $S_L, S_H > 0$ forces $Q_A \leq 0$. The single exceptional chamber (a -pattern 12|0, b -pattern 01|2) carries the closed-form parametrization $a_2 = a_3 = p$, $a_1 = p + 1 + u$, $b_1 = b_2 = q$, $b_3 = q + 1 + v$, in which

$$Q_A = -\frac{p+q+1}{2} R, \quad R = p^2 + 2pq - pu - pv + 3p + q^2 - qu - qv + 3q - 2uv - 4u - 4v,$$

and the exact expansion of $S_L + S_H$ is at most -2 when $u + v \geq 1$ and strictly negative when $u = v = 0$, $p + q \geq 1$. Hence $S_L > 0$, $S_H > 0$ forces $p = q = u = v = 0$, where $Q_A = 0$. The full certificate table and its exact-coefficient audit are proof object (i) of Section 14. $\square$

Together, $Q_A > 0$ and $Q_A \leq 0$ are the contradiction: the cell A-CH is empty.

11 The V and Λ shapes

Let the defect order be V : $p < q$, $p < r$, $q \parallel r$, with windows $I_p = [A, B]$, $I_q = [C_0, D]$, $I_r = [E, F]$ (siblings exchanged so $D \leq F$; realizability gives $A \leq C_0$, $A \leq E$, $B \leq D$, $E \leq D$). The feasible states and weights are

$$\Gamma = \{(i, j, k) \in I_p \times I_q \times I_r : i \leq j, i \leq k\}, \quad W(i, j, k) = 1 + \mathbf{1}_{\{j=k\}}.$$

Theorem 11.1 (Sibling coincidence; V-D empty). *Under closed-third avoidance, the sibling heavy gaps coincide: $\rho_q = \rho_r = D$. Consequently the all-distinct pattern is impossible for V .*

Proof. The sibling marginal $M_q(j) = \sum_{i \in I_p, i \leq j} |I_r \cap [i, m]| + \mathbf{1}_{\{j \in I_r\}} |I_p \cap (-\infty, j]|$ is non-decreasing on I_q , so its unique heavy gap is the right endpoint D . The marginal M_r is nondecreasing through D ; beyond D its first term is constant and its diagonal term vanishes, while at D the diagonal contributes $|I_p| > 0$, so $M_r(D) > M_r(k)$ for all $k > D$. Uniqueness pins $\rho_r = D = \rho_q$. $\square$

Theorem 11.2 (V-CH is vacuous). *No realizable V -configuration has a common heavy gap.*

Proof. For $i \in I_p$ put $Q_i = |I_q \cap [i, m]|$, $R_i = |I_r \cap [i, m]|$, $T_i = |I_q \cap I_r \cap [i, m]|$; fixing i , the feasible (j, k) are the two suffixes and the diagonal contributes one extra extension, so the marginal mass of p is $M_p(i) = Q_i R_i + T_i$ — a product of suffix counts, hence *nonincreasing* in i . The marginal of q , $M_q(j) = \sum_{i \in I_p, i \leq j} |I_r \cap [i, m]| + \mathbf{1}_{\{j \in I_r\}} |I_p \cap (-\infty, j]|$, is *nondecreasing* on I_q (both terms are; the diagonal indicator switches on at E and stays on through $D \leq F$). Now suppose h is heavy for all three defects. If $h < D$, then the upper tail of q at h contains $M_q(D) \geq M_q(h) > N/3$, contradicting the tail bound of Theorem 8.4; so $h = D$. If $h > A$, the lower tail of p at h contains $M_p(A) \geq M_p(h) > N/3$; so $h = A$. Hence $A = D$, and realizability's $A \leq C_0 \leq D$ forces $C_0 = D$: the window $I_q = \{D\}$ is a singleton, contradicting Lemma 8.3. $\square$

Theorem 11.3 (Two-equal partition). *The exactly-two-equal dispatch for the three shapes with relations partitions into 14 cells: 5 for V , 5 for Λ , 4 for Q . Eight are closed by the coincidence and alignment exclusions above and their duals; the survivors are the tied-siblings V cell ($\rho_q = \rho_r > \rho_p$), its Λ dual ($\rho_q = \rho_r < \rho_p$), and four Q cells treated in Section 12.*

In the tied-siblings cell the defect p 's marginal decreases in i (larger i shrinks the feasible (j, k) -region), so $\rho_p = A$, the left endpoint of I_p . The two pinned quantities are the strict heavy-gap tails

$$H = \#\{\text{weighted states with } i > A\}, \quad L = \#\{\text{weighted states with } j < D\},$$

and avoidance gives $3H < N$ and $3L < N$. The cell is emptied by:

Theorem 11.4 (Full-Cone TE(2=1) Theorem). *Every canonical realizable tied-siblings instance satisfies $N \leq 3 \max(H, L)$.*

Proof by certificate corpus. Canonical sibling exchange gives $D \leq F$, splitting the cone into the disjoint facets $D < F$ and $D = F$. Translating D to 0, realizability confines the endpoint parameters to a four-variable integer cone whose weak orderings give exactly 31 chambers on each facet (of the 75 ordered weak partitions of four labels, 31 satisfy $x > y$). On the strict facet, each chamber carries two exact signed certificates (62 in total) expressing the branch identities $3H - N = \alpha(N - 3L) + \beta(N - 3H) + R_H$ and $3L - N = \alpha(N - 3L) + \beta(N - 3H) + R_L$ with $\alpha, \beta \geq 0$ rational and residuals coefficientwise nonnegative, which force $N \leq 3 \max(H, L)$ chamber by chamber. On the boundary facet $D = F$ the stronger inequality holds: all 31 residual identities $3L - N = R_L$ have coefficientwise nonnegative R_L (391 nonzero monomials, minimum nonzero coefficient $1/2$), so $N \leq 3L$ outright. The full-cone partition of the realizable domain into the two facets is exhaustive and disjoint. Certificates, generators, and independent replays are proof objects (ii)–(iv) of Section 14. $\square$

Since $\max(H, L) \geq N/3$ contradicts both strict tails, V-2E is empty — and with it the last V cell. The Λ shape now falls by an exact duality.

Theorem 11.5 (Reflection transfer; Λ empty). *Reflection of chain gaps by $R_m(x) = m - x$ (reversing the chain) bijects the feasible Λ states $\{(i, j, k) : j \leq i, k \leq i\}$ with the canonical V states, preserves the fiber weight $1 + \mathbf{1}_{\{j=k\}}$ and hence N , maps windows $[a, b]$ to $[m-b, m-a]$, exchanges strict lower and upper tails, and sends every same-label orientation numerator S to $N - S$ — so closed-third avoidance, realizability, and heavy-gap data transfer exactly.*

Consequently: Λ -D is empty by Theorem 11.1, Λ -CH by Theorem 11.2, and the tied-siblings Λ cell maps onto the V TE(2=1) cone, where Theorem 11.4 contradicts the two transferred strict guards. All of Λ is empty.

12 The Q shape

Let the defect order be Q : $x < y$, z incomparable to both. The feasible states are $(i, j, k) \in I_x \times I_y \times I_z$ with $i \leq j$, weighted $W = 1 + \mathbf{1}_{\{i=k\}} + \mathbf{1}_{\{j=k\}}$ (the full diagonal $i = j = k$ has weight 3).

Theorem 12.1 (Q-D empty). *Under closed-third avoidance, pairwise-distinct heavy gaps are impossible: the three interval chambers $b < c$, $b = c$, $c < b$ of the comparable pair's windows $[a, b]$, $[c, d]$ exhaust the legal systems, and each either forces coincidence or degeneracy of heavy indices or yields a genuine defect-chain cut whose orientation numerator lies strictly inside $(N/3, 2N/3)$.*

Theorem 12.2 (Q-CH empty). *If x, y, z share a heavy gap r , then one of the strict tails $L_x(r)$, $H_y(r)$ is the orientation numerator of a genuine incomparable defect-chain pair and satisfies $N/3 < T < 2N/3$; the complementary side's heavy atom ($> N/3$) supplies the upper bound.*

Theorem 12.3 (Q1, Q2 empty). *In the two two-equal cells whose tied pair is the comparable pair $\{x, y\}$, the equal heavy gaps $\rho_x = \rho_y$ are excluded structurally: the Q state set has no constraint $i \leq k$, and the exact unique-mode separation of the x and y marginals forces $\rho_x \neq \rho_y$.*

The remaining cells Q3, Q4 tie z with one of the comparable defects: Q3 is $\rho_x = \rho_z < \rho_y$ (tied $\{x, z\}$, y 's heavy gap above), Q4 is $\rho_x < \rho_y = \rho_z$. Write $X = [a, b]$, $Y = [c, d]$, $Z = [e, f]$ for the windows, with lengths $n_x, n_y, n_z \geq 2$ (Lemma 8.3). The comparable pair's windows meet in exactly one of three chambers, and the exact marginal mode analysis of the banked classification disposes of each: if $b = c$ (one-point overlap), a nonsingleton Z creates a two-point maximal plateau in the tied marginals, so no *unique* heavy gap exists and the cell is infeasible; if $b < c$ (*disjoint*), the cell is killed by Lemma 12.4 below; if $c < b$ (*genuine overlap*), the tied coincidence forces the endpoint chamber ($Z = [e, c]$, $X = Y = [c, b]$ for Q3; $Z = [b, f]$, $X = Y = [c, b]$ for Q4), which transfers to the guarded- V cone of Section 11 with a one-unit correction (Theorems 12.5 and 12.6).

Lemma 12.4 (Disjoint-branch closure). *In the disjoint chamber $b < c$, both cells are impossible under closed-third avoidance.*

Proof. Here mode uniqueness forces $Z = [b, c]$ with $\rho_x = b$, $\rho_y = c$, and the tie determines the cell: $\rho_z = b$ (cell Q3) exactly when $n_y > n_x$, and $\rho_z = c$ (cell Q4) exactly when $n_x > n_y$. The exact extension count is $N = n_z n_x n_y + n_x + n_y$. For Q3 select the genuine chain pair $\{x, c_b\}$ (eligible since $a < b$); its orientation numerator is the strict lower tail of x at its heavy gap,

$$A = (n_x - 1)(n_z n_y + 1), \quad 3A - N = n_z n_y (2n_x - 3) + 2n_x - n_y - 3,$$

which is positive for $n_y > n_x \geq 2$: at $n_x = 2$ it equals $(n_z - 1)n_y + 1 > 0$, and for $n_x \geq 3$ it is at least $n_y(4n_x - 7) + 2n_x - 3 > 0$. The complementary mass contains the heavy atom, $N - A \geq$

$\mu_x(b) > N/3$, so $A < 2N/3$. Thus $N/3 < A < 2N/3$, contradicting avoidance. For Q4, symmetrically, the pair $\{c_{c+1}, y\}$ (eligible since $c < d$) has numerator $A = (n_y - 1)(n_z n_x + 1)$, the strict upper tail of y at c , with $3A - N = n_z n_x (2n_y - 3) + 2n_y - n_x - 3 > 0$ for $n_x > n_y \geq 2$ by the same two-case check, and $A < 2N/3$ from the heavy atom again. $\square$

Theorem 12.5 (Corrected full-diagonal transfer). *For every realizable corrected Q3-L or Q4-R overlap instance, gap reversal (Q3-L) or direct relabeling (Q4-R) identifies the Q state set with a canonical guarded- V state set, and the fiber weights agree except at the full diagonal, where Q carries weight 3 against V 's 2. The exact count relations are*

$$N_Q = N_V + 1, \quad H_Q = H_V + 1, \quad L_Q = L_V,$$

with the extra unit on a single named sibling orientation.

Theorem 12.6 (Q3, Q4 empty). *Under closed-third avoidance the corrected overlap branches are impossible.*

Proof from the certificate atlases. Failure of both corrected bounds means $N_Q > 3H_Q$ and $N_Q > 3L_Q$; by Theorem 12.5 and integrality this is equivalent to

$$N_V - 3H_V \geq 3 \quad \text{and} \quad N_V - 3L_V \geq 0.$$

If $H_V \geq L_V$, Theorem 11.4 gives $3H_V \geq N_V$, contradicting the first inequality. If $L_V > H_V$, the theorem gives $3L_V \geq N_V$ and only equality $N_V = 3L_V$ survives; the certificates exclude it. On strict-facet chambers whose residual R_L has a positive constant term, equality is immediate; on the zero-constant chambers the signed identity $3L_V - N_V = \alpha(N_V - 3L_V) + \beta(N_V - 3H_V) + R_L$ with $\beta > 0$ evaluates, at $N_V = 3L_V$ and $L_V > H_V$, to a strictly positive right side against a zero left side. On the boundary facet the equality-face audit gives the strict floor $3L_V - N_V \geq 1$ on every realizable point. Integrality then supplies the missing unit in each case, including the tied case $L_V = H_V + 1$, where $H_Q = L_Q = L_V$ and $N_Q = N_V + 1 \leq 3L_V = 3 \max(H_Q, L_Q)$. Finally the *aligned* family ($A = C_0$, $B = D = E$, $D < F$) admits closed forms with $n = D - A + 1 \geq 2$, $m = F - D + 1 \geq 2$:

$$N_Q = \frac{mn(n+1)}{2} + n + 1, \quad H_Q = \frac{mn(n-1)}{2} + n, \quad L_Q = \frac{mn(n-1)}{2},$$

whence $3H_Q - N_Q = mn(n-2) + 2n - 1 \geq 3$ directly. The corrected chamber-by-chamber and boundary certificates, with their independent replay, are proof object (v) of Section 14. $\square$

13 The Three-Defect Chain Theorem

Theorem 13.1 (Three-Defect Chain Theorem). *Every finite ordinal-sum-indecomposable non-chain poset P with a maximum chain C and $|P \setminus C| \leq 3$ has an incomparable pair $\{u, v\}$ with $1/3 \leq \Pr(u \prec v) \leq 2/3$.*

Proof. Assume closed-third avoidance. If $|P \setminus C| \leq 2$, Theorem 4.1 already yields a balanced pair (defect 0 contradicts non-chain; defect 1 has width two). Let $|P \setminus C| = 3$. If the defects form a chain, Proposition 9.2 applies. Otherwise, by Proposition 9.1 the defect order is (up to duality) A, V, Λ , or Q , and by Theorem 8.4 the heavy pattern is D, 2E,

or CH. Every cell of the resulting grid is empty: A-D and A-2E by Theorem 10.1, A-CH by Theorem 10.2; V-D by Theorem 11.1, V-CH by Theorem 11.2, V-2E by Theorems 11.3 and 11.4; all of Λ by Theorem 11.5; Q-D by Theorem 12.1, Q-CH by Theorem 12.2, Q-2E by Theorem 12.3 (cells Q1, Q2), Lemma 12.4 (the disjoint and one-point chambers of Q3, Q4), and Theorem 12.6 (their overlap chambers). The contradiction hypothesis is untenable, so some incomparable pair lands in the closed middle third. The dependency walk is acyclic: each cell closure consumes only the rigidity theory of Section 8, the model of Section 9, and its own certificates. $\square$

Corollary 13.2 (Defect at most three). *Every finite non-chain poset of defect at most three has an incomparable pair $\{u, v\}$ with $1/3 \leq \Pr(u \prec v) \leq 2/3$.*

Proof. Chains of the canonical factorization $P = P_1 \oplus \cdots \oplus P_k$ (Theorem 9.3) concatenate: a maximum chain of P is the concatenation of maximum chains of the factors, so the defect of P is the sum of the defects of the factors. Since P is not a chain, some indecomposable factor P_i is not a chain, and its defect is between 1 and 3; Theorem 13.1 gives a balanced pair in P_i , and Theorem 9.3 preserves its orientation probabilities in P . $\square$

14 Proof objects and verification

The chamber theorems above are proved by finite families of *exact symbolic identities*: on each chamber, the quantities N, H, L (or N, L_1, H_3, Q_A) are polynomials with rational coefficients in the chamber's nonnegative integer coordinates, and each certificate is an identity between such polynomials whose validity is checked coefficientwise in exact rational arithmetic — no floating point, no sampling, no interpolation in the checker. The corpora are machine-readable artifacts; each was re-derived by an *independent replay* (a separately written checker that reconstructs the identities from the raw definitions), and the artifacts are anchored by SHA-256 hashes. The load-bearing corpora are:

- (i) **CH atlas** (Theorem 10.2): 169 weak-order chambers, 168 exact (α, β) certificates plus the exceptional-chamber identity; exact coefficient audit of every residual. Decision artifact:
9eb6031fecf65abb51f16ed73adc8d166bb3582ecd94467f4415220f8d94a87e
- (ii) **TE(2=1) strict-facet atlas** (Theorem 11.4): 31 chambers, 62 signed certificates; replay executed 49,152 exact identity-component checks and 142,125 numeric spot checks. Atlas / exact coefficient data:
3541901a46c6e9603933d95546d3adb075df15570e6cc6134c7a67ed2c8777a7
59e2b841ce2cb653d900a73092165f1b78547f4bf6023c4b04adf677a878174d
- (iii) **TE(2=1) boundary-facet atlas** ($D=F$): 31 chambers, 391 nonnegative residual monomials, equality-face audit with minimum residual 1 on all $\{0, 1\}$ -faces and 2,304 finite-envelope rows. Atlas / exact coefficient data:
e5cdeb27fb431b3e0ef84c6edbc95dd2e525585ec665278cf67564aab6ab3f219
7aad8e7378dfe5475ba1dc3b8283e8250adf64daffe3359a0a0460b23c6288a5a
- (iv) **Full-cone replay and Λ reflection replay** (Theorems 11.4, 11.5): fresh checkers partitioned 6,048 canonical rows exhaustively into 5,292 strict and 756 boundary rows

with no gap or overlap, and verified the reflection on 173,613 interval triples, 6,355,899 tail identities, and 3,751,704 chain-cut identities, plus actual linear-extension duality on all realizable cases through $m = 3$. Replay record / reflection log:

```
6fce15f9b70e629b175f0175a07fe76f69fab7b0c118ee431491f219271bca2
a9c266e2bd909c33368a10602d4588ea874c8d6c5c02de44ad53633feef0e193
```

- (v) **Corrected Q replay** (Theorem 12.6): 31 strict chambers, 62 signed certificates, 31 boundary identities, 7,776 strict and 2,304 boundary coverage rows, no negative coefficient and no failed identity. Replay record:

```
58c6d5909596794f2561882c625afb3193d077b92f9918dc0fce656a0cbd1e8e
```

All hashes are SHA-256 of the artifact files exactly as published. The complete corpora — certificate tables, exact rational coefficient data, generators, independent checkers, and a `SHA256SUMS` manifest covering every file — accompany this preprint in the public artifact repository github.com/emios-labs/emi-papers (directory `defect-chains`, tagged release matching this draft), together with a supplementary document assembling the full banked derivations behind the statement-level theorems (Theorems 10.1, 11.3, 12.1, 12.2, 12.3, and 12.5). The replays corroborate and gate the symbolic corpora; the theorems rest on the exact identities themselves, which a referee can re-expand from the published artifacts with any computer algebra system.

15 Discussion: the defect ladder ahead

The reduction routes one might hope for are provably blocked at three defects: no representation as a two-defect chain exists inside a connected three-defect core, the canonical ordinal-sum factorization produces no usable factor there, and deletion fails quantitatively (restriction of the uniform distribution to $P - u$ is not uniform after reinsertion). The direct architecture of Sections 8–13 — avoidance rigidity, an exhaustive case grid, and exact certificate corpora for the surviving cells — was forced by these obstructions, and it is the contribution we expect to scale.

Three of its pillars are already defect-count-free: the strong-order theorem, heavy-gap rigidity, and the Helly coincidence mechanism of Theorem 10.1 apply verbatim to any number of pairwise-incomparable defects, and canonical factorization reduces the general conjecture to indecomposable posets. What is not yet defect-count-free is the endgame: at k defects the shape classification grows, and the common-heavy antichain cell needs a k -free replacement for Theorem 10.2. The concrete open target is a *weighted-gap split-tail inequality*: for antichain defects x, y sharing the heavy gap under closed-third avoidance, bound $|\Pr(G_x < G_y) - \Pr(G_x > G_y)|$ from the joint weighted fiber structure. Marginals alone provably do not suffice — independent marginals with below-tail $1/3$ /heavy $2/3$ against heavy $2/3$ /above-tail $1/3$ reach ≈ 0.78 — so any proof must use the tied-diagonal weights and the coupling through the remaining defects. A bound of $1/3$ for some pair would collapse the entire arbitrary- k antichain slice at once.

Other open questions: the defect-four ladder (new shapes, including the first width-four cores); whether any aligned member of Section 6 attains balance constant exactly $1/3$ (cf. Theorem 5.1); and the interval-order connection — incomparable-defect k -defect chains are

interval orders with k nontrivial intervals, so the antichain slice is a first approximation to the open interval-order case.

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